

Math 1232 Summer 2025

Mastery Quiz 8

Due Tuesday, July 29

This mastery quiz has two topics. Everyone should do M3, but if you already have a 2/2 on S6, feel free to skip it. Please do your best, but don't worry if you make a minor error—your goal is to demonstrate your mastery of the underlying material.

Feel free to consult your notes, but please don't discuss the actual quiz questions with other students in the course.

Remember that you are trying to demonstrate that you understand the concepts involved. For all these problems, justify your answers and explain how you reached them. Do not just write “yes” or “no” or give a single number.

Please turn in this quiz in class on Tuesday. You may print this document out and write on it, or you may submit your work on separate paper; in either case make sure your name is clearly on it. If you absolutely cannot turn it in in person, you can submit it electronically but this should be a last resort.

Topics on this quiz:

- Major Topic 3: Series Convergence
- Secondary Topic 6: Sequences and Series

Name: Solutions

Major Topic 3: Series Convergence

Remember to show *all* your work—this includes stating any convergence tests you use, as well as computing any limits that arise.

- (a) Analyze the convergence of the series $\sum_{n=2}^{\infty} \frac{n}{\ln(n)}$.

Solution. By L'Hôpital's rule, we compute that

$$\lim_{n \rightarrow \infty} \frac{n}{\ln(n)} = \lim_{x \rightarrow \infty} \frac{x}{\ln(x)} \stackrel{\text{L'H}}{=} \lim_{x \rightarrow +\infty} \frac{1}{1/x} = \lim_{x \rightarrow \infty} x = \infty.$$

Since this isn't zero, the series diverges by the divergence test. This is a positive series, so this is the same as saying it converges absolutely.

- (b) Analyze the convergence of the series $\sum_{n=1}^{\infty} \frac{n}{n^4 + 1}$.

Solution. We compute

$$\begin{aligned} \int_1^{\infty} \frac{x}{x^4 + 1} dx &= \lim_{t \rightarrow \infty} \int_1^t \frac{x}{x^4 + 1} dx \\ &= \lim_{t \rightarrow \infty} \frac{1}{2} \arctan(x^2) \Big|_1^t = \lim_{t \rightarrow \infty} \frac{1}{2} (\arctan(t^2) - \arctan(1)) \\ &= \frac{1}{2} (\pi/2 - \pi/4) = \frac{\pi}{8}. \end{aligned}$$

This is finite, hence the improper integral converges. Therefore by the integral test, the series $\sum_{n=1}^{\infty} \frac{n}{n^4+1}$ converges. This is a positive series, so this is the same as saying it converges absolutely.

Alternatively (**and probably this is easier**), we could observe that $\frac{n}{n^4+1} \leq \frac{n}{n^4} = \frac{1}{n^3}$. Since $\sum_{n=1}^{\infty} \frac{1}{n^3}$ converges by the p -series test, we can conclude that $\sum_{n=1}^{\infty} \frac{n}{n^4+1}$ converges by the comparison test.

(c) Analyze the convergence of the series $\sum_{n=1}^{\infty} \frac{4n^3 + 1}{n^4 + n + 3}$.

Solution. Really any time we have a rational function, easiest will be the Limit Comparison Test. The numerator and denominator have degrees 3 and 4, respectively, so we compare this to $n^3/n^4 = 1/n$. Since

$$\lim_{n \rightarrow \infty} \frac{\frac{4n^3+1}{n^4+n+3}}{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{4n^4 + n}{n^4 + n + 3} = 4$$

is finite and nonzero, the Limit Comparison Test says this series has the same behavior as the series $\sum_{n=1}^{\infty} \frac{1}{n}$, which we know diverges by the p -series test. Therefore $\sum_{n=1}^{\infty} \frac{4n^3+1}{n^4+n+3}$ diverges.

We could also have used the integral test: we have

$$\begin{aligned} \int_1^{\infty} \frac{4x^3 + 1}{x^4 + x + 3} dx &= \lim_{t \rightarrow \infty} \int_1^t \frac{4x^3 + 1}{x^4 + x + 3} dx = \lim_{t \rightarrow \infty} \ln(x^4 + x + 3) \Big|_1^t \\ &= \lim_{t \rightarrow \infty} \ln(t^4 + t + 3) - \ln(5) = \infty \end{aligned}$$

Where we've done the u -substitution $u = x^4 + x + 3$ to find the antiderivative. The improper integral diverges, so by the integral test the series $\sum_{i=1}^{\infty} \frac{4n^3+1}{n^4+n+3}$ diverges.

Secondary Topic 6: Sequences and Series

(a) Compute $\sum_{n=1}^{\infty} \frac{2}{n^2 + 4n + 3}$. Rigorously justify your computation of this limit.

Solution. We can do a partial fractions decomposition: we have

$$\begin{aligned} 2 &= A(n+1) + B(n+3) \\ 2 &= 2B & \Rightarrow B &= 1 \\ 2 &= -2A & \Rightarrow A &= -1. \end{aligned}$$

Then our partial sums are

$$\begin{aligned} s_k &= \sum_{n=1}^k \frac{2}{n^2 + 4n + 3} = \sum_{n=1}^k \frac{1}{n+1} - \frac{1}{n+3} \\ &= \left(\frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \left(\frac{1}{4} - \frac{1}{6}\right) + \cdots + \left(\frac{1}{k+1} - \frac{1}{k+3}\right) \\ &= \frac{1}{2} + \frac{1}{3} - \frac{1}{k+2} - \frac{1}{k+3}. \end{aligned}$$

Thus

$$\sum_{n=1}^{\infty} \frac{2}{n^2 + 4n + 3} = \lim_{k \rightarrow \infty} \frac{1}{2} + \frac{1}{3} - \frac{1}{k+2} - \frac{1}{k+3} = \frac{1}{2} + \frac{1}{3} = \frac{5}{6}.$$

(b) Consider the sequence $(a_n) = (6, 2, \frac{2}{3}, \frac{2}{9}, \dots)$. Find a formula for the n th term a_n . Compute $\lim_{n \rightarrow \infty} a_n$.

Solution. $a_n = \frac{18}{3^n}$ or $a_n = 2 \cdot 3^{2-n}$. We have

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{18}{3^n} = 0.$$

We can justify this by saying the bottom goes to infinity.

(c) Compute $\sum_{n=1}^{\infty} \frac{2^{2n+1}}{3^n}$.

Solution. This is a geometric series with $a = \frac{8}{3}$ and $r = \frac{4}{3}$. Since $r > 1$ this series does not converge, and the sum of the series is ∞ .